

HW0414

2026-04-16

8.16

```
library(readxl)
```

```
## Warning: 套件 'readxl' 是用 R 版本 4.2.3 來建造的
```

```
vacation <- read_excel("C:/finance ecomatrics/data/vacation.xlsx")  
View(vacation)
```

```
library(lmtest)
```

```
## Warning: 套件 'lmtest' 是用 R 版本 4.2.3 來建造的
```

```
## 載入需要的套件：zoo
```

```
## Warning: 套件 'zoo' 是用 R 版本 4.2.3 來建造的
```

```
##  
## 載入套件：'zoo'
```

```
## 下列物件被遮斷自 'package:base':  
##  
##      as.Date, as.Date.numeric
```

```
library(sandwich)  
library(dplyr)
```

```
## Warning: 套件 'dplyr' 是用 R 版本 4.2.3 來建造的
```

```
##  
## 載入套件：'dplyr'
```

```
## 下列物件被遮斷自 'package:stats':  
##  
##      filter, lag
```

```
## 下列物件被遮斷自 'package:base':  
##  
##      intersect, setdiff, setequal, union
```

a.

```
mod_ols <- lm(miles ~ income + age + kids, data = vacation)
summary_ols <- summary(mod_ols)
ci_a <- confint(mod_ols, "kids", level = 0.95)

cat("--- (a) OLS Results ---\n")
```

```
## --- (a) OLS Results ---
```

```
print(summary_ols$coefficients["kids", ])
```

```
##      Estimate   Std. Error    t value   Pr(>|t|)
## -81.826418749  27.129596044  -3.016131114  0.002899464
```

```
cat("95% CI for KIDS (a): [", ci_a[1], ",", ci_a[2], "]\n\n")
```

```
## 95% CI for KIDS (a): [ -135.3298 , -28.32302 ]
```

b.

```
par(mfrow = c(1, 2))
plot(vacation$income, resid(mod_ols), main="Residuals vs Income", xlab="Income", ylab="Residuals", pch=20)
abline(h=0, col="red", lty=2)
plot(vacation$age, resid(mod_ols), main="Residuals vs Age", xlab="Age", ylab="Residuals", pch=20)
abline(h=0, col="red", lty=2)
```



```
par(mfrow = c(1, 1))
```

Residuals vs Income :

Observed Pattern: There is a distinct pattern widening to the right. At lower income levels, the residuals are tightly clustered around zero. As income increases, the vertical spread of the residuals increases significantly.

This indicates that the error variance is not constant but increases with INCOME. This is a clear visual confirmation that heteroskedasticity is present in the model.

Residuals vs Age :

Observed Pattern: The data points form a relatively uniform horizontal band. The vertical spread of the residuals remains fairly consistent (roughly between -1000 and 1000) across all age groups.

This suggests that the error variance does not systematically change with AGE, which is consistent with the assumption of homoskedasticity regarding this specific variable.

Conclusion: Heteroskedasticity is definitely present in the data, and it is systematically related to INCOME rather than age.

C.

```
df_sorted <- vacation %>% arrange(income)

# GQ Test
gq_result <- gqtest(mod_ols, order.by = ~income, data = vacation, fraction = 20)
cat("--- (c) Goldfeld-Quandt Test ---\n")
```

```
## --- (c) Goldfeld-Quandt Test ---
```

```
print(gq_result)
```

```
##
## Goldfeld-Quandt test
##
## data:  mod_ols
## GQ = 3.1041, df1 = 86, df2 = 86, p-value = 1.64e-07
## alternative hypothesis: variance increases from segment 1 to 2
```

```
cat("\n")
```

H_0 : Homoskedasticity (the variance of the errors is constant across all income levels).

GQ Statistic = 3.1041: This F-statistic is essentially the ratio of the residual variance in the high-income segment to the residual variance in the low-income segment. A value of roughly 3.1 tells us that the variance among high-income households is more than 3 times larger than that of low-income households.

p-value = 1.64e-07, which is virtually zero.

Conclusion: Since the p-value is small and substantially less than the standard significance level of $\alpha = 0.05$, we reject the null hypothesis. We have overwhelming statistical evidence of heteroskedasticity; specifically, the error variance strictly increases with income (as stated in the alternative hypothesis).

d.

```
mod_robust <- coeftest(mod_ols, vcov = vcovHC(mod_ols, type = "HC1"))
# CI: coef +/- t_crit * robust_se
se_robust <- mod_robust["kids", "Std. Error"]
coef_kids <- mod_robust["kids", "Estimate"]
t_crit <- qt(0.975, df = mod_ols$df.residual)
ci_d_low <- coef_kids - t_crit * se_robust
ci_d_up <- coef_kids + t_crit * se_robust

cat("--- (d) Robust SE Results ---\n")
```

```
## --- (d) Robust SE Results ---
```

```
cat("95% Robust CI for KIDS (d): [", ci_d_low, ",", ci_d_up, "]\n\n")
```

```
## 95% Robust CI for KIDS (d): [ -139.323 , -24.32986 ]
```

Comparing (a) and (d), the confidence interval widened from 107 to 115. Because traditional OLS assumes constant variance. When heteroskedasticity is present, OLS typically underestimates the true standard errors, providing an artificially narrow and overconfident interval. The Robust SE in part (d) corrects this flaw. It honestly accounts for the extreme noise introduced by high-income households, resulting in a wider, but statistically valid and reliable, confidence interval.

e.

```
# weight= 1 / (Income^2)
mod_gls <- lm(miles ~ income + age + kids, data = vacation, weights = 1/(income^2))
ci_e_conv <- confint(mod_gls, "kids", level = 0.95)

# Robust GLS
mod_gls_robust <- coeftest(mod_gls, vcov = vcovHC(mod_gls, type = "HC1"))
se_gls_robust <- mod_gls_robust["kids", "Std. Error"]
ci_e_rob_low <- coef_kids_gls <- mod_gls_robust["kids", "Estimate"] - t_crit * se_gls_robust
ci_e_rob_up <- mod_gls_robust["kids", "Estimate"] + t_crit * se_gls_robust

cat("--- (e) GLS Results ---\n")
```

```
## --- (e) GLS Results ---
```

```
cat("95% Conv. GLS CI for KIDS: [", ci_e_conv[1], ",", ci_e_conv[2], "]\n")
```

```
## 95% Conv. GLS CI for KIDS: [ -119.8945 , -33.71808 ]
```

```
cat("95% Robust GLS CI for KIDS: [", ci_e_rob_low, ",", ci_e_rob_up, "]\n")
```

```
## 95% Robust GLS CI for KIDS: [ -121.4134 , -32.19919 ]
```

the Conventional GLS interval—its width shrank dramatically to 86.17, making it the narrowest and most precise estimate among all methods. OLS treats every observation equally. However, we know high-income households are extremely noisy (high variance). GLS applies Weighted Least Squares (WLS), giving more weight to the reliable, low-variance observations (low income) and less weight to the noisy, high-variance ones. By optimally re-weighting the data, GLS achieves maximum efficiency (it is BLUE), yielding the tightest possible confidence interval without losing validity.

The slight widening in Robust GLS compared to Conv. GLS simply means our theoretical weighting assumption wasn't 100% perfect, so the robust correction adds a final layer of safety.